

WOOSEOP HWANG

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Education

University of Oxford

Oct 2023 – Jun 2024

Master of Mathematics and Theoretical Physics (MMathPhys)

- First Class Honours (GPA 4.0 equivalent).
- Master's Dissertation: *Preparing Ground and Excited States Using Adiabatic CoVaR*.

University of Oxford

Oct 2020 – Jun 2023

Bachelor of Arts in Physics (BA)

- First Class Honours (GPA 4.0 equivalent), with Distinction.

Experience

Qraft Technologies

Aug 2024 – Present

AI/Quant Researcher (Alternative Military Service)

Seoul, South Korea

Reinforcement-Learning Order Execution

- Built an optimized model-training environment on a Rust/Python simulator.
- Researched PPO-based deep reinforcement-learning models for order execution.
- Developed a Hindsight Learning-based reward function, improving credit assignment and training efficiency.
- Refined the model architecture and state encoder to enhance the learned state representation.

Domestic-Market ETF Arbitrage

- Built a C++ high-frequency-trading (HFT) backtesting and trading platform.
- Improved ETF fair-value computation and the order-execution system, increasing the capacity of existing arbitrage strategies by over 100%.
- Introduced a real-time bond-pricing algorithm, expanding the arbitrage universe.

Data cybernetics ssc GmbH

Oct 2023 – Mar 2024

Researcher

Remote, Germany

Applications of an In-House Quantum State Preparation Algorithm (Q-Alchemy)

- Applied a low-rank approximation that relaxes the state-preparation step for encoding data into quantum circuits, across several application problems.
- QCNN — analyzed conditions under which approximate encoding sharply reduces circuit depth while preserving classification performance.
- HHL — analyzed how input-stage approximation error is amplified in the final solution as a function of the problem's condition number.
- Portfolio optimization — showed that the low-rank structure of the return covariance makes the components truncated by approximation mostly noise, reducing state-preparation cost without degrading solution quality.

Yonsei University

Dec 2022 – Aug 2023

Student Researcher

Seoul, South Korea

Helstrom-Classifer Research

- Implemented a quantum-inspired classifier that efficiently evaluates the Helstrom-measurement-operator classifier — optimal for discriminating non-orthogonal quantum states — even for multiple data copies.
- Benchmarked against hyperparameter-tuned classical ML classifiers across diverse datasets.

Publications

Preparing Ground and Excited States Using Adiabatic CoVaR 2024 – 2025

First-author, with Prof. Bálint Koczor (University of Oxford)

- Developed a hybrid quantum-classical algorithm for preparing ground and excited eigenstates of target Hamiltonians, addressing the warm-start limitation of CoVaR via adiabatic Hamiltonian morphing. Demonstrated on three settings: ground and excited states of a spin-ring Hamiltonian; combinatorial optimization via ground-state preparation (MAX-CUT encoded as an Ising Hamiltonian); and excited states of the Schwinger model, a high-energy-physics toy model. Published in *New Journal of Physics* (2025).

Generalized Gaussian Temporal Difference Error for Uncertainty-aware Reinforcement Learning 2024

Co-author, with S. Kim, J. Lee, N. Cho, S. Han (Qraft Technologies)

- Proposed a generalized-Gaussian model of temporal-difference error capturing higher-order moments (kurtosis), improving the estimation of aleatoric and epistemic uncertainty in deep reinforcement learning. arXiv:2408.02295 (preprint).

Quantum-Inspired Classification via Efficient Simulation of Helstrom Measurement 2023 – 2024

First-author, with Prof. Daniel K. Park (Yonsei), I. F. Araujo, C. Blank

- Derived an efficient simulation of the Helstrom Quantum Centroid classifier for arbitrary data-copy counts, revealing that copy number is a hyperparameter rather than a monotonic improvement and enabling datasets larger than the underlying Hilbert space dimension. arXiv:2403.15308 (preprint).

Awards & Honors

Best Student Poster Award · Asia Quantum Information Science Conference (AQIS) 2024

Fitzgerald Prize · Exeter College, University of Oxford 2021, 2024

East Scholarship · Exeter College, University of Oxford 2020–2024

Gold Medal · South East Asia Mathematics Competition 2020

Bronze Medal · Singapore Mathematics Olympiad 2020

Talks & Presentations

Poster Presentation · Asia Quantum Information Science Conference (AQIS), Quantum Techniques in Machine Learning (QTML) 2024

Skills

Programming: Python (Advanced), C++ (Intermediate), Rust (Intermediate), Qiskit, PennyLane, $\LaTeX$

Areas: Quantum Algorithms, Combinatorial Optimization, Numerical Analysis & Simulation, Machine Learning / Reinforcement Learning, Quantitative Research

Languages: English (Native), Korean (Native)